

Case Study: Architecting a High-Throughput Hybrid Document AI Platform

1. Business Scenario & Mission Overview

We are spearheading a high-stakes initiative to deploy a **serverless, API-driven Document AI platform** on AWS. The system is designed for the heavy-duty asynchronous processing of massive PDF volumes, specifically targeting complex, multi-page invoices, legal contracts, and corporate reports.

The platform must provide seamless **Optical Character Recognition (OCR)**, granular entity extraction, and **LLM-powered insights**. As an AIOps Engineer, your objective is to design an infrastructure that reconciles high-speed parallel processing with extreme cost efficiency and operational stability.

Mission Statement: To engineer a serverless, event-driven architecture capable of processing documents up to **500+ pages** with **sub-5-minute end-to-end latency**, maintaining a cost profile of less than **\$0.10 per document** while ensuring **99.9% system availability**.

2. Document Specifications & Throughput Targets

The following technical boundaries define the mission's operational envelope. Candidates must architect specifically to address the risks identified below.

Table 1: System Constraints & Engineering Risks

Attribute	Target Specification	Engineering Challenges / Risk Vectors
Document Length	Up to 1,000 pages per doc	Lambda timeouts; requires sophisticated chunking.

File Size	Up to 300 MB per upload	Ingestion bottlenecks; requires S3 Transfer Acceleration.
Document Types	Invoices, Contracts, Forms	Layout complexity risks inconsistent extraction accuracy.
Daily Throughput	10,000+ docs (Sustained)	Requires elastic resource management for state-machines.
Burst Capacity	50,000 docs (Peak)	Risk of hitting service quotas; requires async decoupling.
Latency (Large)	< 5 minutes (P95)	Enforces non-blocking models; avoids synchronous waits.

3. Functional Requirements: The AI Pipeline

The platform must execute a standardized **7-stage lifecycle**. Candidates must address these "Required Pipeline Capabilities" in their design:

1. **Ingestion:** Validate upload, generate unique Job ID, and persist primary file in S3.
 2. **Pre-processing:** Segment large PDFs into 20–50 page chunks; detect orientation and enhance image quality.
 3. **OCR & Structure Extraction:** Execute asynchronous extraction (Text/Forms/Tables) via managed services.
 4. **Merging:** Consolidate individual chunk results into a single, ordered document structure.
 5. **AI Extraction:** Apply **Small Language Models (SLMs)** or **LLMs** for classification and entity recognition.
 6. **Post-processing:** Validate results against schemas, normalize data, and structure output into JSON.
 7. **Storage:** Persist final results to S3 and update persistent job status in the database.
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4. Technical Challenges & Problem Set

4.1 Latency & Scaling

Achieve sub-5-minute completion for 1,000-page documents while navigating the 128K token limit of LLM context windows. Design an event-driven pattern using SQS/EventBridge to protect downstream AI services from throttling during 5x peak bursts.

4.2 Cost & Reliability

Target <\$0.10 per 100-page document. Define a comprehensive **idempotency strategy** and a "poison pill" mechanism to handle failed jobs via automated replay and manual inspection workflows.

4.3 Specialized Challenge: Managed vs. Self-Hosted Inference

The business is considering a transition to **Self-Hosted LLMs** (e.g., Llama 3 or Mistral) hosted on **Amazon EKS** with **GPU orchestration**.

- **Orchestration:** Describe how you would manage GPU-based autoscaling (e.g., Karpenter or KEDA).
- **Cost Modeling:** Provide a break-even analysis: at what volume does a dedicated GPU cluster (vLLM/TGI) become cheaper than Bedrock's token-based pricing?
- **Observability:** Define how the observability stack (Prometheus/Grafana) tracks **GPU utilization** versus Managed Model metrics.

5. MLOps, Security & Compliance Mandates

Operational Standards

- **IaC:** Mandatory use of **AWS CDK with Python** for all stacks.
- **CI/CD:** GitHub Actions requiring **80% unit test coverage** and CDK synthesis validation.
- **Model Eval:** Tracking framework for Precision, Recall, and F1 scores.
- **Deployment:** Support for **10% traffic canary rollouts** over a 1-hour soak period.

Security & Observability

- **Distributed Tracing:** End-to-end tracing across the stack using **AWS X-Ray**.
- **PII Protection:** Automatic detection/masking of sensitive data (SSNs, emails) in logs and outputs.
- **Data Sovereignty:** Encryption at rest via **KMS (CMK)** and 365-day data retention policies.
- **Metrics:** Real-time tracking of token usage, **Time To First Token (TTFT)**, and cost per job.

6. Candidate Deliverables

Candidates must produce the following four technical artifacts:

1. **Proposed Architecture:** Detailed description of serverless/event-driven components and GPU orchestration layers.
2. **Operational Strategy:** Plan for monitoring system health (P95 latency, GPU metrics, and cost per job).
3. **Security & Compliance Posture:** Description of alignment with **GDPR** and **SOC 2** requirements.
4. **Risk Mitigation Plan:** Identify mitigations for service quotas, cost overruns, and model hallucinations.

7. Glossary of Terms

- **IDP:** Intelligent Document Processing
- **SLM/LLM:** Small (<10B) and Large (>10B) Language Models
- **TTFT:** Time To First Token
- **vLLM:** High-throughput LLM serving library
- **DLQ:** Dead Letter Queue